



# SIMATS Online Programming Lab

## JAVA PROGRAMMING



MD Imthiaz I  
192521360RD

### Exception- Practice- CO3

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#### QUESTIONS

**Q1**

Exception  
10 marks

**Q2**

IllegalArgumentException  
10 marks

**Q3**

user-defined exception named  
WeakPasswordException.  
10 marks

**Q4**

Email Validation  
10 marks

**Q5**

Banking Transaction Using  
Exception Propagation  
10 marks



5/5 answered

#### Q1 Exception

10 marks

Write a Java program to store five integers in an array. Accept an index from the user and display the corresponding element. Handle invalid index access.

Test Case

Input

Array: 10 20 30 40 50

Enter Index:

6

Output

Exception: Invalid array index.

#### Your Code

```
1 import java .util.Scanner;
2 public class Main{
3     public static void main(String[] args)
4     {
5         try{
6             Scanner sc=new Scanner(System.in);
7             int[] arr=new int[5];
8             for (int i=0;i<5;i++){
9                 arr[i]=sc.nextInt();
10            }
11            int index=sc.nextInt();
12            System.out.println(arr[index])
13        }
```

```
14         catch(Exception e) {  
15             System.out.println("Exception:  
16         }  
17  
18     }  
19 }
```

### Input

```
10 20 30 40 50  
6
```

### Output

```
Exception: Invalid array index.
```

### Visible Test Cases

Test Case 1



Test Case 2



Test Case 3

**10.00 / 10 marks** **Submitted**



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**Q5**

Banking Transaction Using  
Exception Propagation  
10 marks



5/5 answered

#### Q2 IllegalArgumentException

10 marks

##### Problem Statement

Write a Java program to accept marks (0–100) for a student. If the entered mark is less than 0 or greater than 100, throw an IllegalArgumentException and display an appropriate message.

##### Sample Input

120

##### Sample Output

Exception: Invalid marks. Marks should be between 0 and 100.

#### Your Code

```
1 import java.util.Scanner;
2 public class Main{
3     public static void main(String[] args)
4     {
5         Scanner sc=new Scanner(System.in);
6         int marks=sc.nextInt();
7         try{
8             checkMarks(marks);
9             System.out.println("Valid mark");
10        }catch(IllegalArgumentException e)
11        {
12            System.out.println("Exception: "+e.getMessage());
13        }
14    }
15    static void checkMarks(int marks){
16        if (marks<0 || marks>100){
```

```
15         throw new IllegalArgumentExceptionExce  
16     }  
17  
18     }  
19 }
```



### Input

120

### Output

Exception: Invalid marks. Marks should be  
between 0 and 100.

### Visible Test Cases

Test Case 1



Test Case 2



Test Case 3



10.00 / 10 marks

 Submitted



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**Q1**

Exception  
10 marks

**Q2**

IllegalArgumentException  
10 marks

**Q3**

user-defined exception named  
WeakPasswordException.  
10 marks

**Q4**

Email Validation  
10 marks

**Q5**

Banking Transaction Using  
Exception Propagation  
10 marks



5/5 answered

#### Problem Statement

Write a Java program to accept a password. If the password contains fewer than 8 characters, throw a user-defined exception named WeakPasswordException.

#### Sample Input

java123

#### Sample Output

Exception: Password must contain at least 8 characters.

#### Your Code

```
1 import java.util.Scanner;
2 public class Main {
3     public static void main(String[] args)
4     {
5         Scanner sc=new Scanner(System.in);
6         String password=sc.next();
7         try{
8             checkPassword(password);
9             System.out.println("Password : ");
10        }
11        catch(WeakPasswordException e) {
12            System.out.println("Exception: ");
13        }
```

```
14     static void checkPassword(String passw
15         if(password.length()<8){
16             throw new WeakPasswordExceptio
17         }
18     }
19 }
20 class WeakPasswordException extends Except
21     public WeakPasswordException(String me
22         super(message);
23     }
24 }
```



### Input

java123

### Output

Exception: Password must contain at least 8 characters.

### Visible Test Cases

Test Case 1



Test Case 2



Test Case 3



10.00 / 10 marks

 Submitted



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### QUESTIONS

**Q1**

Exception  
10 marks

**Q2**

IllegalArgumentException  
10 marks

**Q3**

user-defined exception named  
WeakPasswordException.  
10 marks

**Q4**

Email Validation  
10 marks

**Q5**

Banking Transaction Using  
Exception Propagation  
10 marks



5/5 answered

### Q4 Email Validation

10 marks

#### Problem Statement

Write a Java program to accept an email address. If the email does not contain the '@' symbol, throw a user-defined exception named InvalidEmailException.

#### Sample Input

student@gmail.com

#### Sample Output

Exception: Invalid email address.

### Your Code

```
1 import java.util.Scanner;  
2 public class Main{  
3     static void validateEmail(String email){  
4         if(!email.contains("@")){  
5             throw new InvalidEmailException;  
6         }  
7     }  
8     public static void main(String[] args){  
9         Scanner sc=new Scanner(System.in);  
10        String email=sc.next();  
11        try{  
12            validateEmail(email);  
13            System.out.println("Valid email address");  
14        }  
15        catch(InvalidEmailException e){  
16            System.out.println("Exception: Invalid email address");  
17        }  
18    }  
19 }
```

```
17         }
18     }
19 }
20 class InvalidEmailException extends Exception
21     public InvalidEmailException(String message) {
22         super(message);
23     }
24 }
```



### Input

stufent@gmail.com



### Output

Exception: Invalid email address.

### Visible Test Cases

Test Case 1



Test Case 2



Test Case 3



10.00 / 10 marks

✓ Submitted





## SIMATS Online Programming Lab

### JAVA PROGRAMMING



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**Q2**

IllegalArgumentException  
10 marks

**Q3**

user-defined exception named  
WeakPasswordException.  
10 marks

**Q4**

Email Validation  
10 marks

**Q5**

Banking Transaction Using  
Exception Propagation  
10 marks



5/5 answered

#### Problem Statement

Write a Java program to simulate a banking transaction using three methods:

withdraw()

validateBalance()

processTransaction()

If the withdrawal amount exceeds the account balance, throw an exception from processTransaction(). Allow it to propagate through the other methods and handle it only in main().

#### Sample Input

Balance:

5000

Withdraw:

7000

#### Sample Output

Exception propagated to main().

InsufficientBalanceException: Withdrawal amount exceeds available balance.

## Your Code

```
1 import java.util.Scanner;
2
3 public class Main {
4     public static void main(String[] args)
5     {
6         Scanner sc = new Scanner(System.in);
7         int balance = sc.nextInt();
8         int withdrawAmount = sc.nextInt();
9
10        try {
11            withdraw(balance, withdrawAmount);
12        } catch (InsufficientBalanceException e) {
13            System.out.println("Exception: " + e.getMessage());
14            System.out.println("Insufficient balance");
15        }
16    }
17
18    static void withdraw(int balance, int amount) {
19        validateBalance(balance, amount);
20    }
21
22    static void validateBalance(int balance, int amount) {
23        processTransaction(balance, amount);
24    }
25
26    static void processTransaction(int balance, int amount) {
27        if (amount > balance) {
28            throw new InsufficientBalanceException("Insufficient balance");
29        } else {
30            System.out.println("Transaction successful");
31        }
32    }
33
34    class InsufficientBalanceException extends Exception {
35        public InsufficientBalanceException(String message) {
36            super(message);
37        }
38    }
```



## Input

5000

7000



## Output

```
Exception propagated to main().  
InsufficientBalanceException: Withdrawal  
amount exceeds available balance.
```

## Visible Test Cases

Test Case 1



Test Case 2



Test Case 3

**0.00 / 10 marks** Run Save Submit

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